

Krishna Anand

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GitHub: <https://github.com/KANAND6031>

EDUCATION

Indian Institute of Information Technology, Manipur

B.Tech in Computer Science and Engineering (2023-2027)

CGPA: 8.7 (Till 5th Sem)

R.P.S Public School

12th, CBSE Board (2022)

89%

PTJM Saraswati Vidya Mandir, Rajgir

10th, CBSE Board (2020)

91%

TECHNICAL SKILLS

Languages: C, C++, Python, HTML, CSS, JavaScript, PHP, React

Database: MySQL

Technologies/Frameworks: Linux/Ubuntu, Google Colab, GitHub

Concepts: Data Structures and Algorithms, Object-Oriented Programming, Database Management System, Operating System, Computer Networks

PROJECTS

Chest X-Ray Classification for Multi-Disease Detection

Python, PyTorch, TensorFlow/Keras, Grad-CAM, Google Colab

Feb 2026 – Apr 2026

- Designed and trained a custom 4-block CNN from scratch in PyTorch (~3.3M parameters) to classify chest X-rays into 4 categories: COVID-19, Normal, Pneumonia, and Tuberculosis — achieving **92% test accuracy**, outperforming the fine-tuned ResNet50 baseline (89%).
- Implemented a 3-phase fine-tuning strategy for ResNet50 (frozen base → top-10 layer unfreezing → final refinement) using TensorFlow/Keras with class-weighted cross-entropy loss to handle severe TB class imbalance (~700 vs 4000+ Pneumonia samples).
- Implemented Gradient-weighted Class Activation Mapping (Grad-CAM) on both models to generate heatmap overlays, confirming clinically relevant activations such as peripheral opacities and lower-lobe consolidations in COVID-19 and Pneumonia cases.
- Extended Grad-CAM analysis by dividing the lung into six anatomical zones for automated localization of the highest-activation region per prediction, improving clinical interpretability.
- Applied data augmentation (random horizontal flip, rotation $\pm 10^\circ$, color jitter) via torchvision transforms; trained the custom CNN for 80 epochs with Adam optimizer and Cosine Annealing LR scheduling, achieving TB recall of 0.94.

OS Scheduling Simulator

GitHub Repository

- Built a CPU Scheduling Simulator implementing FCFS, SJF, SRTF, Round Robin, LJF, and LRTF algorithms in C++.
- Built an interactive frontend using React.js to display dynamic Gantt charts and waiting time, turnaround time, and CPU utilization.
- Implemented core scheduling algorithms in C++ for efficient computation and optimized performance.
- Improved understanding of operating system concepts through clear visualization.

Sorting Algorithm Visualizer

HTML, CSS, JavaScript, Sorting Algorithm

GitHub Repository

- Developed an interactive web application to visualize sorting algorithms including Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, and Quick Sort.
- Implemented dynamic array visualization with animation, allowing users to observe step-by-step sorting operations.
- Added features such as custom array input, random array generation, start/pause/resume controls, and speed adjustment.

ACHIEVEMENTS

- NPTEL Gold Certification- In Business Intelligence and Analytics.
- NPTEL Silver Certification- In Introduction to Information Retrieval.
- Secured 1st place in the Code Hunt competition at the college Tech Fest.
- Secured 3rd place in the Robo War competition at the college Tech Fest.
- NPTEL Certification- In Getting Started with Competitive Programming.
- Participated in Smart India Hackathon 2024.

CODING PROFILE

- Good understanding of Array, String, Stack, Recursion, Queue, Linked List.
- Solved 150+ Data Structures and Algorithms problems across multiple coding platforms.
- LeetCode: <https://leetcode.com/u/ucOpv8RaMc/>

EXTRA-CURRICULAR ACTIVITIES

- Managed and organized technical events during the college Tech Fest.
- Participated in Coding events at college.
- Competed in inter-house badminton competition.